

ACCESS CONTROL LISTS

Standard, Extended, Port, and VLAN ACLs on IOS-XE

A Segmented Campus Lab: RACL, PACL, and VACL Interoperation

Platform: Cisco IOS-XE vIOS (routers) / IOSvL2 (switch) on EVE-NG

Lab Series: Network Engineering Lab Manuals

Revision Date: July 2026

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1. Lab Objectives

This lab builds a working mental model of Cisco access control by implementing every major ACL type on a single, coherent topology and then proving — with live traffic — how they interact when more than one is present at once. By the end of this lab you will be able to:

- Configure and apply standard numbered ACLs (1–99) to filter traffic based on source address alone.
- Configure and apply extended numbered ACLs (100–199) to filter on source, destination, protocol, and port.
- Convert both ACL types to named format and understand why named ACLs are preferred in production.
- Configure a Port ACL (PACL) on a Layer 2 access port and understand its ingress-only, hardware-level scope.
- Configure a VLAN ACL (VACL) using vlan access-map / vlan filter and understand its VLAN-wide scope.
- Explain — and verify on the wire — the precedence order when a Router ACL (RACL), PACL, and VACL are all present in the same traffic path.

PLATFORM NOTE — Platform used for this lab

R1 and R-ISP run Cisco IOS-XE vIOS (routed ACL behavior — RACL, standard/extended, numbered/named).

SW1 runs Cisco IOSvL2, the Layer 2/3 switch image for EVE-NG. IOSvL2 supports Port-ACLs and VLAN Access Maps (VACLs) in its documented feature set, so this lab is achievable entirely in EVE-NG without a hardware Catalyst.

As with every lab in this series, verify actual CLI output against your own EVE-NG instance — IOSvL2 is a software forwarding path, not real TCAM, so hardware-only verification commands (e.g. show platform ... acl counters) will not exist here. Where that matters it is called out explicitly.

2. Lab Topology

Six devices, one edge, one distribution switch, three VLANs. The topology is deliberately small so that every ACL type can be exercised without unrelated complexity — the goal is to isolate what each ACL type does and where it sits in the packet's path, not to build a large network.

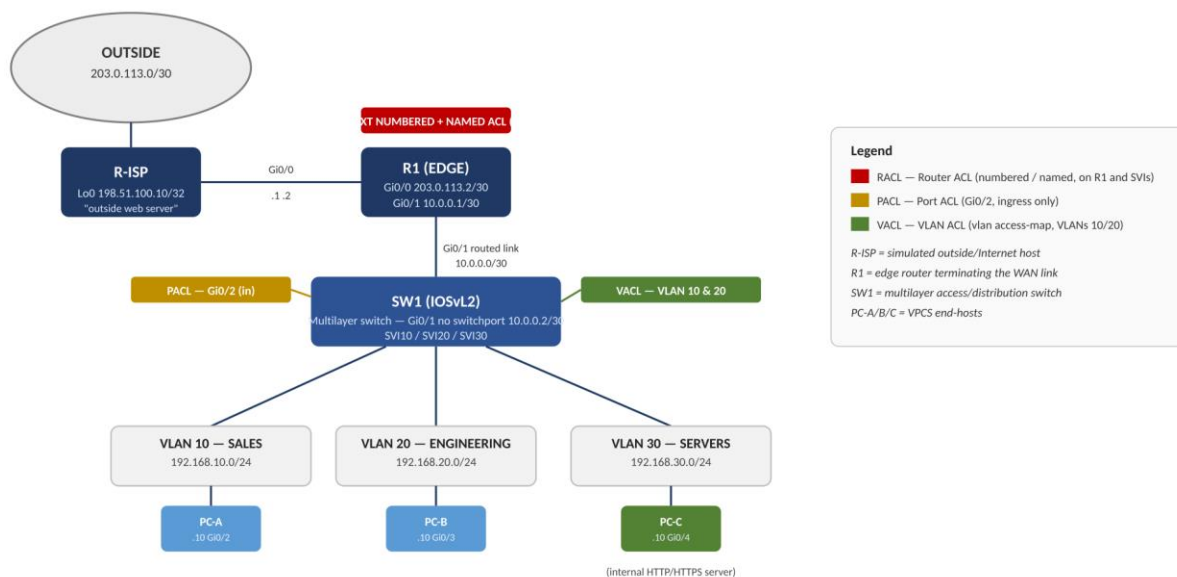


Figure 1. ACL lab topology — RACL on R1, PACL on Gi0/2, VACL on VLANs 10/20.

2.1 Device Roles

Device	Role	Platform / EVE-NG Image	ACL Types Hosted
R-ISP	Simulated outside network / Internet host	vios-adventerprise9 (IOS-XE vIOS)	None (traffic source only)
R1	Edge / gateway router terminating the WAN link	vios-adventerprise9 (IOS-XE vIOS)	Standard & Extended, Numbered & Named (RACL)
SW1	Multilayer distribution/access switch	viosl2-adventerprise9 (IOSvL2)	RACL (SVIs), PACL (Gi0/2), VACL (VLANs 10/20)
PC-A	Sales end host	VPCS	n/a
PC-B	Engineering end host	VPCS	n/a
PC-C	Internal server (simulated HTTP/HTTPS)	VPCS	n/a

2.2 IP Addressing Plan

Device	Interface	IP Address	Notes
R-ISP	Gi0/0	203.0.113.1/30	Link to R1 (outside)
R-ISP	Lo0	198.51.100.10/32	Simulated public web server
R1	Gi0/0	203.0.113.2/30	Outside interface — inbound extended ACL applied here
R1	Gi0/1	10.0.0.1/30	Routed link to SW1 — inside interface
SW1	Gi0/1	10.0.0.2/30	no switchport — routed uplink to R1
SW1	VLAN 10 (SVI)	192.168.10.1/24	Sales gateway
SW1	VLAN 20 (SVI)	192.168.20.1/24	Engineering gateway
SW1	VLAN 30 (SVI)	192.168.30.1/24	Servers gateway
PC-A	eth0	192.168.10.10/24	Gateway 192.168.10.1, Gi0/2 access VLAN 10
PC-B	eth0	192.168.20.10/24	Gateway 192.168.20.1, Gi0/3 access VLAN 20
PC-C	eth0	192.168.30.10/24	Gateway 192.168.30.1, Gi0/4 access VLAN 30 — internal server

PLATFORM NOTE — Why a routed link instead of a trunk between R1 and SW1

Keeping the R1–SW1 link as a point-to-point routed link (no switchport on SW1's Gi0/1) means all inter-VLAN routing, and every RACL, lives on SW1's SVIs rather than being split across two devices. This keeps the RACL/VACL/PACL interoperability story in Section 8 confined to a single box, which is exactly how you'd read it in Cisco's own documentation on the subject.

3. Base Configuration

Bring the topology up with routing and reachability working end-to-end before any ACL is applied. Every ACL section that follows assumes this baseline is already in place and passes the verification pings at the end of this section.

3.1 R-ISP

```
hostname R-ISP
!
interface Loopback0
 ip address 198.51.100.10 255.255.255.255
!
interface GigabitEthernet0/0
 ip address 203.0.113.1 255.255.255.252
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 203.0.113.2
```

3.2 R1 (Edge Router)

```
hostname R1
!
interface GigabitEthernet0/0
 description OUTSIDE - to R-ISP
 ip address 203.0.113.2 255.255.255.252
 no shutdown
!
interface GigabitEthernet0/1
 description INSIDE - to SW1
 ip address 10.0.0.1 255.255.255.252
 no shutdown
!
ip route 0.0.0.0 0.0.0.0 203.0.113.1
ip route 192.168.10.0 255.255.255.0 10.0.0.2
ip route 192.168.20.0 255.255.255.0 10.0.0.2
ip route 192.168.30.0 255.255.255.0 10.0.0.2
!
line vty 0 4
 transport input ssh
 login local
```

3.3 SW1 (Multilayer Switch)

Create the VLANs, assign access ports, configure SVIs, and route toward R1. Because SW1 is doing inter-VLAN routing, ip routing must be enabled globally.

```
hostname SW1
ip routing
!
vlan 10
 name SALES
vlan 20
 name ENGINEERING
```

```

vlan 30
  name SERVERS
!
interface GigabitEthernet0/1
  description ROUTED UPLINK - to R1
  no switchport
  ip address 10.0.0.2 255.255.255.252
  no shutdown
!
interface GigabitEthernet0/2
  description ACCESS - PC-A (Sales)
  switchport mode access
  switchport access vlan 10
  no shutdown
!
interface GigabitEthernet0/3
  description ACCESS - PC-B (Engineering)
  switchport mode access
  switchport access vlan 20
  no shutdown
!
interface GigabitEthernet0/4 GigabitEthernet1/0
  description ACCESS - PC-C (Server)
  switchport mode access
  switchport access vlan 30
  no shutdown
!
interface Vlan10
  ip address 192.168.10.1 255.255.255.0
!
interface Vlan20
  ip address 192.168.20.1 255.255.255.0
!
interface Vlan30
  ip address 192.168.30.1 255.255.255.0
!
ip route 0.0.0.0 0.0.0.0 10.0.0.1

```

3.4 End Hosts (VPCS)

```

# PC-A
ip 192.168.10.10/24 192.168.10.1

# PC-B
ip 192.168.20.10/24 192.168.20.1

# PC-C
ip 192.168.30.10/24 192.168.30.1

```

3.5 Baseline Verification

Before touching any ACL, confirm full reachability in both directions:

- PC-A, PC-B, and PC-C can all ping each other and their default gateways.
- PC-A/B/C can ping R1's Gi0/0 (203.0.113.2) and R-ISP's Lo0 (198.51.100.10).

- R-ISP can ping into all three internal VLANs (192.168.10.10 / .20.10 / .30.10).

```
PC-A> ping 198.51.100.10
84 bytes from 198.51.100.10 icmp_seq=1 ttl=253 time=3.412 ms
84 bytes from 198.51.100.10 icmp_seq=2 ttl=253 time=2.877 ms
84 bytes from 198.51.100.10 icmp_seq=3 ttl=253 time=2.901 ms
```

VERIFY — Do not proceed until this passes

Every ACL section below is a controlled restriction applied on top of full reachability. If baseline connectivity is broken before you apply an ACL, you won't be able to tell whether a later 'deny' is the ACL working as intended or a routing problem underneath it.

4. Standard Numbered ACL (RACL)

Standard ACLs match on source address only — nothing about destination, protocol, or port. Range: 1–99 and 1300–1999 (expanded range). Because they carry so little context, standard ACLs should be applied as close to the destination as possible, and the classic textbook use case is exactly what we'll do here: restrict management access to a router's own VTY lines.

4.1 Scenario

Only hosts in the Servers/Management VLAN (192.168.30.0/24) should be permitted to open a Telnet or SSH session to R1. Everyone else — including Sales and Engineering — should be rejected before the login prompt even appears.

4.2 Configuration — R1

```
R1(config)# access-list 10 remark Permit mgmt subnet to VTY only
R1(config)# access-list 10 permit 192.168.30.0 0.0.0.255
R1(config)# access-list 10 deny any log
!
R1(config)# line vty 0 4
R1(config-line)# access-class 10 in
```

CAUTION — access-class vs access-group

Standard ACLs applied to VTY lines use access-class, not access-group. access-group is reserved for physical/logical interfaces. This is a common source of a silent misconfiguration — the ACL compiles fine either way, but access-group under a line configuration is simply not a valid command, and access-class under an interface is likewise invalid.

4.3 Verification

```
PC-B> ping 192.168.30.10
! PC-B (Engineering) attempts SSH to R1 outside gateway - denied

R1# show access-lists 10
Standard IP access list 10
 10 permit 192.168.30.0, wildcard bits 0.0.0.255 (4 matches)
 20 deny any log (2 matches)
```

The match counters on line 20 confirm Engineering's connection attempts were dropped by the implicit-deny-turned-explicit-deny-with-log line, which also lets you see the denies in the log rather than guessing from an absent counter.

4.4 Why Standard ACLs Are Placement-Sensitive

Because a standard ACL only evaluates the source address, filtering early in the path (near the source) risks blocking traffic that was actually destined somewhere the sender should legitimately reach. Placing standard ACL 10 directly on R1's VTY lines — the actual resource being protected — avoids that ambiguity entirely; there's only one thing it can mean.

5. Extended Numbered ACL (RACL)

Extended ACLs match on source, destination, protocol, and Layer 4 port information. Range: 100–199 and 2000–2699 (expanded range). Because they can express a precise policy, extended ACLs should be applied as close to the source as possible — the opposite placement rule from standard ACLs — so unwanted traffic is dropped before it consumes any bandwidth downstream.

5.1 Scenario

R1's outside interface (Gi0/0) faces the simulated Internet. The only traffic that should be allowed in from outside is HTTP/HTTPS destined for the internal server (PC-C, 192.168.30.10) and ICMP echo-replies so troubleshooting still works. Everything else inbound from outside — including any attempt to reach directly into the Sales or Engineering VLANs — is dropped.

5.2 Configuration — R1

```
R1(config)# access-list 100 remark Inbound filter on outside interface
R1(config)# access-list 100 permit tcp any host 192.168.30.10 eq 80
R1(config)# access-list 100 permit tcp any host 192.168.30.10 eq 443
R1(config)# access-list 100 permit icmp any any echo-reply
R1(config)# access-list 100 permit icmp any any unreachable
R1(config)# access-list 100 deny ip any any log
!
R1(config)# interface GigabitEthernet0/0
R1(config-if)# ip access-group 100 in
```

NOTE — Direction matters more here than on a standard ACL

This ACL is applied inbound on the outside interface, so it only ever evaluates traffic arriving from R-ISP toward the inside. Traffic initiated from inside (Sales/Engineering/Servers going out to R-ISP) is never checked against ACL 100 — it's a different direction on the same interface, and outbound direct-initiated traffic is unaffected by this policy.

5.3 Verification

```
R-ISP# telnet 192.168.30.10 80
Trying 192.168.30.10, 80 ... Open

R-ISP# ping 192.168.10.10
```

Success rate is 0 percent (0/5)

```
R1# show access-lists 100
Extended IP access list 100
 10 permit tcp any host 192.168.30.10 eq www (3 matches)
 20 permit tcp any host 192.168.30.10 eq 443
 30 permit icmp any any echo-reply (5 matches)
 40 permit icmp any any unreachable
 50 deny ip any any log (1 match)
```

Port 80 to the server succeeds; an attempt to reach Sales directly from outside is silently dropped and picked up by the logging deny at the bottom.

5.4 Numbered ACL Ranges (Reference)

Range	Type	Notes
1–99	Standard IP (original)	Source address only
100–199	Extended IP (original)	Source, destination, protocol, port
1300–1999	Standard IP (expanded)	Used once the original range is exhausted
2000–2699	Extended IP (expanded)	Used once the original range is exhausted

6. Standard Named ACL (RACL)

Functionally identical to a numbered standard ACL, but referenced by name instead of number. Named ACLs are strongly preferred in production because the name is self-documenting, entries can be inserted or removed by sequence number without rebuilding the whole list, and there's no need to track which numbers are already in use elsewhere in the network.

6.1 Scenario

Engineering (VLAN 20) should not be able to source traffic toward Sales (VLAN 10) at all — a simple, source-only separation policy between the two client VLANs, enforced on SW1.

6.2 Configuration — SW1

```
SW1(config)# ip access-list standard ENG-TO-SALES-DENY
SW1(config-std-nacl)# deny 192.168.20.0 0.0.0.255
SW1(config-std-nacl)# permit any
SW1(config-std-nacl)# exit
!
SW1(config)# interface Vlan10
SW1(config-if)# ip access-group ENG-TO-SALES-DENY out
```

NOTE — Explicit permit any is required

Every ACL — numbered or named, standard or extended — ends in an implicit deny any. Once you add even one explicit line to a named standard ACL, you must add permit any yourself if you don't want the implicit deny to silently catch traffic you meant to allow (in this case, everyone except Engineering reaching Sales).

6.3 Verification

```
PC-B> ping 192.168.10.10
Request timeout.
Request timeout.

PC-C> ping 192.168.10.10
84 bytes from 192.168.10.10 icmp_seq=1 ttl=254 time=1.203 ms

SW1# show ip access-lists ENG-TO-SALES-DENY
Standard IP access list ENG-TO-SALES-DENY
 10 deny 192.168.20.0, wildcard bits 0.0.0.255 (6 matches)
 20 permit any (11 matches)
```

7. Extended Named ACL (RACL)

Named extended ACLs combine the granularity of extended matching with the readability of a name. This is the format you'll actually write in production; numbered extended ACLs mostly show up in legacy configs or quick one-off tests.

7.1 Scenario

Both client VLANs (Sales and Engineering) should reach the internal server on HTTP, HTTPS, and DNS, and should be able to ping it for troubleshooting — but nothing else. This is enforced once, centrally, on the SVI that faces the Servers VLAN, rather than duplicated on each client SVI.

7.2 Configuration — SW1

```
SW1(config)# ip access-list extended CLIENTS-TO-SERVERS
SW1(config-ext-nacl)# permit tcp 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255 eq 80
SW1(config-ext-nacl)# permit tcp 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255 eq 443
SW1(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 eq 80
SW1(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 eq 443
SW1(config-ext-nacl)# permit udp any host 192.168.30.10 eq 53
SW1(config-ext-nacl)# permit icmp any 192.168.30.0 0.0.0.255 echo
SW1(config-ext-nacl)# deny ip any 192.168.30.0 0.0.0.255 log
SW1(config-ext-nacl)# permit ip any any
SW1(config-ext-nacl)# exit
!
SW1(config)# interface Vlan30
SW1(config-if)# ip access-group CLIENTS-TO-SERVERS out
```

NOTE — Applied outbound on the destination SVI, on purpose

This ACL is applied out on Vlan30 rather than in on Vlan10/Vlan20. That means it evaluates all routed traffic destined for the Servers VLAN in one place, regardless of which client VLAN it came from, instead of needing an identical ACL duplicated on every client-facing SVI. Section 8 revisits exactly this inbound-vs-outbound SVI placement as part of the RACL processing order.

7.3 Verification

```
PC-A> curl http://192.168.30.10
<html>...200 OK...</html>

PC-A> ping 192.168.30.10 -p 23
! (SSH/Telnet to the server) Request timeout.

SW1# show ip access-lists CLIENTS-TO-SERVERS
Extended IP access list CLIENTS-TO-SERVERS
 10 permit tcp 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255 eq www (14 matches)
 20 permit tcp 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255 eq 443
 30 permit tcp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 eq www (9 matches)
 40 permit tcp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 eq 443
 50 permit udp any host 192.168.30.10 eq domain
 60 permit icmp any 192.168.30.0 0.0.0.255 echo (10 matches)
 70 deny ip any 192.168.30.0 0.0.0.255 log (3 matches)
 80 permit ip any any
```

8. Port ACL (PACL)

A PACL is a standard or extended IP ACL — or a MAC-extended ACL for non-IP traffic — applied directly to a Layer 2 switchport rather than a routed interface. It is the earliest possible filtering point a frame can hit on the switch: before VLAN bridging, before inter-VLAN routing, before anything else runs. PACLs are ingress-only; there is no hardware support for an outbound PACL.

8.1 Scenario

PC-A (Sales, Gi0/2) should be able to reach its default gateway and everything beyond it as normal, but should not be able to directly reach any other host inside VLAN 10 at Layer 2 — a simple form of client isolation enforced right at the access port, independent of anything configured on the SVIs.

8.2 Configuration — SW1

```
SW1(config)# ip access-list extended PACL-PCA-ISOLATE
SW1(config-ext-nacl)# permit ip host 192.168.10.10 host 192.168.10.1
SW1(config-ext-nacl)# deny ip host 192.168.10.10 192.168.10.0 0.0.0.255
SW1(config-ext-nacl)# permit ip any any
SW1(config-ext-nacl)# exit
!
SW1(config)# interface GigabitEthernet0/2
SW1(config-if)# ip access-group PACL-PCA-ISOLATE in
```

CAUTION — PACL direction and interface type

ip access-group ... in is the only valid direction on a Layer 2 port — attempting out will be rejected by the CLI. A PACL can only be applied to a switchport (access or, in prefer-port mode, trunk); applying one to a routed (no switchport) interface simply makes it a normal RACL instead, with none of the port-level behavior described in Section 10.

8.3 Verification

```
PC-A> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=0.912 ms

PC-A> ping 192.168.30.10
84 bytes from 192.168.30.10 icmp_seq=1 ttl=254 time=1.301 ms
! reaches the server fine - that's routed traffic leaving VLAN 10 through the SVI, not
another VLAN-10 host

SW1# show access-lists PACL-PCA-ISOLATE
Extended IP access list PACL-PCA-ISOLATE
 10 permit ip host 192.168.10.10 host 192.168.10.1 (7 matches)
 20 deny ip host 192.168.10.10 192.168.10.0 0.0.0.255
 30 permit ip any any
```

Note that pinging the server (192.168.30.10) still works — that traffic leaves VLAN 10 immediately via the SVI and is never a VLAN-10-to-VLAN-10 flow, so it never matches the deny line. Only communication with other VLAN-10 hosts is affected, because that traffic is bridged, not routed, and the PACL sees every frame entering Gi0/2 regardless of where it's ultimately headed.

9. VLAN ACL (VACL)

A VACL — configured as a vlan access-map and applied with vlan filter — is not tied to a direction or a specific port at all. It filters every packet that is bridged within the VLAN or routed into/out of the VLAN, on every port that is a member of it. Where a PACL protects one port and an RACL protects one routed interface, a VACL protects the whole VLAN as a unit.

9.1 Scenario

As a blanket security policy, Telnet (TCP/23) should never be usable anywhere inside VLAN 10 or VLAN 20 — not host-to-host, not to the gateway, not routed in from another VLAN — regardless of what any RACL or PACL elsewhere permits. Everything else continues to flow normally.

9.2 Configuration — SW1

```
SW1(config)# ip access-list extended BLOCK-TELNET
SW1(config-ext-nacl)# permit tcp any any eq 23
SW1(config-ext-nacl)# exit
!
SW1(config)# vlan access-map TELNET-FILTER 10
SW1(config-access-map)# match ip address BLOCK-TELNET
SW1(config-access-map)# action drop
SW1(config-access-map)# vlan access-map TELNET-FILTER 20
SW1(config-access-map)# action forward
SW1(config-access-map)# exit
!
SW1(config)# vlan filter TELNET-FILTER vlan-list 10,20
```

CAUTION — The ACL inside a vlan access-map only ever needs permit statements

Match logic inside a VACL sequence only evaluates permit entries in the referenced ACL. A deny entry in that ACL doesn't mean 'forward' — it means 'this sequence didn't match, fall through to the next sequence.' That's why BLOCK-TELNET uses permit tcp any any eq 23 to define what gets dropped, and why a second, empty-

match catch-all sequence (20, action forward) is required so that every other type of traffic — which never matches the permit in sequence 10 — has somewhere to land instead of hitting the implicit deny that ends every vlan access-map.

9.3 Verification

```
PC-A> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=0.884 ms
! ICMP still works fine - only TCP/23 is targeted

PC-A> telnet 192.168.10.1
% Connection timed out; remote host not responding

SW1# show vlan access-map
Vlan access-map "TELNET-FILTER" 10
  Match clauses:
    ip address: BLOCK-TELNET
  Action:
    drop
Vlan access-map "TELNET-FILTER" 20
  Match clauses:
  Action:
    forward
!
SW1# show vlan filter
VLAN Map TELNET-FILTER is filtering VLANs:
 10, 20
```

The VACL is enforced identically at every port in VLAN 10 and VLAN 20 — it doesn't matter whether the Telnet attempt originated at PC-B's own port, was bridged in from another port in the same VLAN, or arrived already routed from a different VLAN. That VLAN-wide, direction-agnostic scope is the defining trait that separates a VACL from both a PACL and an RACL, and it's the subject of the next section.

10. Interoperation of PACL, VACL, and RACL

SW1 now has all three ACL types live at once: a PACL on Gi0/2, a VACL covering VLANs 10 and 20, and RACLs on Vlan10/Vlan20 (from Section 6) and Vlan30 (from Section 7). This section explains, precisely, what order a packet is checked against them — because the answer is not 'whichever is more specific' or 'whichever was configured last.' It's a fixed pipeline defined by where each ACL type physically sits in the switch's forwarding path.

10.1 What Each ACL Type Actually Protects

ACL Type	Applied To	Direction	Scope
PACL	Layer 2 switchport (access or trunk)	Ingress only	One physical port. Filters every frame entering that port — bridged or destined to be routed — before any VLAN or routing decision is made.
VACL	A VLAN, via vlan access-map / vlan filter	Both (not direction-based)	The entire VLAN as a broadcast domain. Filters all traffic bridged within the VLAN and all traffic routed into or out of it, on every port that is a member.

ACL Type	Applied To	Direction	Scope
RACL	A Layer 3 interface (SVI, routed port, sub-interface)	In and/or Out	Only traffic that is actually being routed through that interface. Purely bridged traffic within a VLAN never reaches an RACL at all.

10.2 Overall Filtering Precedence

When a PACL, a VLAN map, and a router ACL are all configured on the same switch, Cisco's IOS-XE security configuration guide defines the precedence as follows:

Traffic Direction	Precedence (highest to lowest)
Ingress (entering the switch)	1. Port ACL → 2. VLAN map (VACL) → 3. Router ACL (RACL)
Egress (leaving the switch)	1. Router ACL (RACL) → 2. VLAN map (VACL) → 3. Port ACL

Notice the order flips between ingress and egress. On the way in, the most physically local check (the port) runs first. On the way out, the most physically local check (the port) runs last — the RACL, being the most 'central' decision, is checked first on egress because it's what decided the packet should leave via that path at all.

10.3 Bridged Traffic (Same VLAN, No Routing Involved)

When PC-A talks to another host in VLAN 10 without ever leaving that VLAN, the SVI is never consulted — the two hosts resolve each other's MAC address directly and the switch simply bridges the frame. An RACL configured on Vlan10 plays no role whatsoever in this flow. The order is:

- 1. Inbound PACL on the ingress switchport (e.g., Gi0/2)
- 2. Inbound VACL on the VLAN
- 3. Outbound VACL on the VLAN, at the egress port

NOTE — This is exactly what Section 8's verification showed

PC-A's Telnet or same-VLAN traffic to another VLAN-10 host is bridged traffic. It hits PACL-PCA-ISOLATE first (Section 8) and, had it survived that, would then hit the VACL TELNET-FILTER (Section 9) — but never any router ACL, because no routing decision is ever made for a purely intra-VLAN flow.

10.4 Routed Traffic (Crossing VLANs)

When PC-A (VLAN 10) talks to PC-C (VLAN 30), SW1 has to route the packet between two SVIs. This is the case where all three ACL types can legitimately be in the path at once, and the order is:

- 1. Inbound PACL on the ingress switchport (Gi0/2)
- 2. Inbound VACL on the ingress VLAN (VLAN 10)
- 3. Inbound RACL on the ingress SVI (Vlan10, if one is applied in)
- 4. Outbound RACL on the egress SVI (Vlan30, if one is applied out)
- 5. Outbound VACL on the egress VLAN (VLAN 30, if one is applied)

There is no step for an outbound PACL — Cisco's switching platforms do not support port ACLs in the egress direction in hardware, which is why every PACL in this lab (and in practice, generally) is written with in as the only option.

10.5 Walking a Real Packet Through SW1

Take a single packet: PC-A (192.168.10.10) sends an HTTP GET to PC-C, the internal server (192.168.30.10). With everything configured in Sections 6–9 active simultaneously, here is exactly what that packet is checked against, in order:

Step	ACL Checked	Outcome for this packet
1	PACL-PCA-ISOLATE (Gi0/2, in)	Source/destination don't match the deny line (destination is not in 192.168.10.0/24) — falls through to permit ip any any. Passes.
2	VACL TELNET-FILTER (VLAN 10, ingress)	Not TCP/23 — doesn't match sequence 10's ACL, falls to sequence 20, action forward. Passes.
3	RACL on Vlan10 (in) — none applied	No inbound RACL configured on Vlan10 in this lab. Skipped.
4	RACL CLIENTS-TO-SERVERS (Vlan30, out)	Matches the TCP/80 permit line for the 192.168.10.0/24 → 192.168.30.0/24 flow. Passes.
5	VACL on VLAN 30 — none applied	No vlan filter configured on VLAN 30 in this lab. Skipped.

Five possible checkpoints, three of which are actually populated in this lab, and the packet has to clear every populated one to reach PC-C. Change any single line in any of those three ACLs and this table is the tool for predicting exactly which step will start dropping the packet — rather than guessing and re-testing from scratch.

10.6 access-group mode: merge vs. prefer port

On platforms that support it, the access-group mode command on a Layer 2 interface controls how a PACL interacts with a VACL and RACL that would otherwise also apply to that traffic.

Mode	Behavior
merge (default)	The PACL, VACL, and RACL entries applicable to the port are logically combined and evaluated together, following the serial order in Sections 10.3/10.4. This is what SW1 has been doing throughout this lab and requires no extra configuration.
prefer port	The PACL alone is applied to ingress traffic on that port; the VACL and any RACL that would otherwise apply are bypassed entirely for traffic entering that port. This mode is required if you need a PACL on a trunk port, and is otherwise a deliberate override for cases where the port-level policy must be the single source of truth.
<pre>SW1(config)# interface GigabitEthernet0/2 SW1(config-if)# access-group mode prefer port ! With this set, TELNET-FILTER (VACL) and any RACL no longer apply ! to traffic entering Gi0/2 - only PACL-PCA-ISOLATE decides its fate.</pre>	
PLATFORM NOTE — IOSvL2 command availability	

Mode	Behavior
	access-group mode is documented on hardware Catalyst platforms (6500/4500/9000 series). Confirm on your own IOSvL2 build whether the command is present before relying on it — if it's absent, the platform is implicitly running in merge mode only, which is the default and the mode used everywhere else in this lab regardless.

10.7 Summary: Choosing Which ACL Type to Use

- Need to isolate or restrict one specific access port, regardless of VLAN policy? → PACL.
- Need a security policy that applies uniformly across an entire VLAN, for both bridged and routed traffic? → VACL.
- Need to control traffic based on routing — between subnets, to/from a specific SVI or WAN interface? → RACL (this is what Sections 4–7 configured).
- Need all three at once? They stack in the fixed order from Sections 10.2–10.4 — design each one assuming the others are also being evaluated, not in isolation.

11. Verification Command Reference

Command	Purpose
<code>show access-lists [name number]</code>	Display all ACLs (numbered and named, standard and extended) with per-line match counters
<code>show ip access-lists [name number]</code>	Same as above, restricted to IP ACLs
<code>show ip interface Gi0/x</code>	Confirm which ACL, if any, is applied inbound/outbound on a Layer 3 interface
<code>show vlan access-map [name]</code>	Display configured vlan access-maps, their sequences, match clauses, and actions
<code>show vlan filter</code>	Display which VLANs a vlan access-map is currently applied to
<code>show access-group mode interface Gi0/x</code>	Confirm merge vs. prefer port mode on a Layer 2 interface (platform-dependent)
<code>show running-config section access-list</code>	Quickly pull every ACL definition out of a large running-config
<code>debug ip packet [detail] (access-list)</code>	Last resort — trace individual packets against a specific ACL; use sparingly, it is CPU-expensive

11.1 A Repeatable Troubleshooting Sequence

- 1. Confirm the base topology is reachable with no ACLs applied (Section 3.5) — rule out routing/VLAN issues first.
- 2. Identify every ACL in the packet's path using the precedence tables in Section 10, not just the one you most recently edited.
- 3. Check match counters with `show access-lists` — an unexpectedly high count on a deny line, or a zero count on a permit line you expected to hit, tells you exactly which ACE is actually being evaluated.
- 4. Remember the implicit deny — an ACL with only deny statements, or one where you forgot the trailing permit, will silently drop everything not explicitly matched.
- 5. For VACLs specifically, remember that only permit entries in the referenced ACL are meaningful match criteria (Section 9.2) — a deny entry inside a vlan access-map's ACL means 'skip to the next sequence,' not 'drop.'

PLATFORM BUG / CORRECTION — Common mistake in this lab series carried over here

As with the OSPF and IPsec labs in this series, always confirm exact IOS-XE syntax on your own image rather than assuming classic IOS documentation applies verbatim — `vlan access-map`, `vlan filter`, and `access-group mode` all have platform-dependent availability and exact output formatting that can differ between `vios`, `IOSvL2`, and real Catalyst hardware.

11.2 Full Configuration Recap

For quick reference, every ACL configured in this lab, in the order it was introduced:

ACL / Map	Type	Applied To	Section
10	Standard numbered (RACL)	R1, line vty 0 4 (access-class in)	4
100	Extended numbered (RACL)	R1, Gi0/0 (ip access-group in)	5
ENG-TO-SALES-DENY	Standard named (RACL)	SW1, Vlan10 (ip access-group out)	6
CLIENTS-TO-SERVERS	Extended named (RACL)	SW1, Vlan30 (ip access-group out)	7
PACL-PCA-ISOLATE	Extended named (PACL)	SW1, Gi0/2 (ip access-group in)	8
TELNET-FILTER	vlan access-map (VACL)	SW1, VLANs 10 & 20 (vlan filter)	9